Presentation Notes: An Introduction to Group Theory

1. The Formal Definition of a Group

Let G be a set of elements and $*$ be a binary operation (a rule for combining any two elements). The pair $(G, *)$ is a **group** if and only if it perfectly satisfies these four axioms:

1. **Closure:** Combining any two elements in the group keeps you in the group.
   (Formal: $\forall a, b \in G, \quad a * b \in G$)
2. **Associativity:** The grouping of operations doesn't matter.
   (Formal: $\forall a, b, c \in G, \quad (a * b) * c = a * (b * c)$)
3. **Identity Element:** There exists a "do nothing" element, usually denoted as e.
   (Formal: $\exists e \in G$ such that $\forall a \in G, \quad e * a = a * e = a$)
4. **Inverse Element:** Every single action has an "opposite" action that undoes it.
   (Formal: $\forall a \in G, \exists a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$)

We let $|G|$ denote the **order** of G which just means the number of elements (cardinality) in the underlying set of G.

2. Group Properties & Proofs

- **Uniqueness of Identity:** There is only *one* "do nothing" action.

  Proof. Suppose a group has two identity elements, e and e'.
  Since e' is an identity, $e * e' = e$.
  Since e is an identity, $e * e' = e'$.
  Therefore, $e = e'$. The identity is strictly unique. □

- **Uniqueness of Inverses:** Each element has exactly *one* unique inverse.

  Proof. Suppose an element a has two inverses, b and c.
  Consider the expression $b * a * c$.
  By associativity, $(b * a) * c = b * (a * c)$.
  Since b is an inverse of a, $(b * a) = e$, so $e * c = c$.

Since c is an inverse of a, $(a * c) = e$, so $b * e = b$.
Therefore, $b = c$. □

- **Cancellation Laws:** We can confidently cancel elements in algebraic equations.

  Proof. Assume $a * b = a * c$.
  Multiply both sides on the left by a^{-1}: $a^{-1} * (a * b) = a^{-1} * (a * c)$.
  By associativity: $(a^{-1} * a) * b = (a^{-1} * a) * c$.
  By definition of inverse: $e * b = e * c$.
  By definition of identity: $b = c$. □

2-1. Examples of Groups

To truly understand what a group is, it helps to see everyday math that perfectly fits the rules, and everyday math that completely fails them.

Two Basic Examples of Groups

1. **The Integers under Addition $(\mathbb{Z}, +)$:** The set of all integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$ with standard addition.
   - *Closure:* Adding two integers always gives another integer.
   - *Identity:* The number 0 (since $a + 0 = a$).
   - *Inverse:* Every number a has a negative $-a$ (since $a + (-a) = 0$).
2. **Non-Zero Rational Numbers under Multiplication $(\mathbb{Q} \setminus \{0\}, \times)$:** The set of all fractions (except zero) with standard multiplication.
   - *Closure:* Multiplying two non-zero fractions gives a non-zero fraction.
   - *Identity:* The number 1 (since $x \times 1 = x$).
   - *Inverse:* Every fraction a/b has a reciprocal b/a (since $\frac{a}{b} \times \frac{b}{a} = 1$).

Two Counterexamples

1. **The Integers under Multiplication $(\mathbb{Z}, \times)$:** *Fails the Inverse Axiom.*
   While it has closure, associativity, and an identity (1), it completely fails the inverse rule. For example, the integer 2 would need an inverse of $1/2$ to equal the identity ($2 \times 1/2 = 1$). However, the fraction $1/2$ is not an integer, so it does not exist in our set!
2. **The Natural Numbers under Addition $(\mathbb{N}, +)$:** *Fails the Identity and Inverse Axioms.*
   If we look at strictly positive whole numbers $\{1, 2, 3, \dots\}$, there is no zero, meaning the group lacks an identity element. Even if we allow zero into our set, it fails the inverse axiom: there are no negative numbers, so the number 5 has nothing in the set it can be added to in order to get back to 0.

3. Subgroups

- **Definition:** A subset H of a group G is a subgroup (written $H \leq G$) if H itself forms a valid group using the exact same operation as G.

The Two-Step Subgroup Test: Let G be a group and let H be a subset of G. H is a subgroup of G ($H \leq G$) if and only if:

1. **Closure under the operation:** For all $a, b \in H$, the combination $a * b$ is also in H.
2. **Closure under inverses:** For every $a \in H$, its inverse a^{-1} is also in H.

Why does this test work? Because the subset H lives inside the parent group G, it automatically inherits associativity. But how do we know the identity element is actually in H?

Proof. For arbitrary $x \in H$, By Step 2 (Closure under inverses), the inverse x^{-1} must also be in H.
By Step 1 (Closure under the operation), combining these two elements must result in an element inside H.
Therefore, $x * x^{-1} \in H$. By definition, $x * x^{-1} = e$.
Thus, $e \in H$. All four group axioms are satisfied! □

Examples of Subgroups:

1. **Even Integers ($2\mathbb{Z} \leq \mathbb{Z}$):** Inside the group of all integers under addition, the even integers form a subgroup. *Check:* 0 is even (Identity). Even + Even = Even (Closure). The negative of an even number is even (Inverses).
2. **Positive Rationals ($\mathbb{Q}^+ \leq \mathbb{Q}^*$):** Inside the group of non-zero fractions under multiplication, the strictly positive fractions form a subgroup. *Check:* 1 is positive. Positive × Positive = Positive. The reciprocal of a positive fraction is still positive.

Applying the Test: An Example

Let's rigorously prove that the multiples of 3 form a subgroup of the integers under addition. Let $G = (\mathbb{Z}, +)$ and let $H = 3\mathbb{Z} = \{3k \mid k \in \mathbb{Z}\}$.

- **Step 0: Is it non-empty?**
  Yes. Let $k = 0$. Since $3(0) = 0$, the identity element 0 is in H.
- **Step 1: Is it closed under addition?**
  Let x and y be arbitrary elements in H.
  By the definition of our set, $x = 3m$ and $y = 3n$ for some integers m, n.
  Add them together: $x + y = 3m + 3n = 3(m + n)$.
  Since $(m + n)$ is just another integer, the result is a multiple of 3. Therefore, $x + y \in H$.

- **Step 2: Is it closed under inverses?**
  Let $x \in H$. So, $x = 3m$ for some integer m.
  In the group of integers, the inverse of x is $-x$.
  $-x = -(3m) = 3(-m)$.
  Since $-m$ is an integer, $-x$ is a multiple of 3. Therefore, $-x \in H$.

Since $3\mathbb{Z}$ passes the Two-Step Test, we can confidently state that $3\mathbb{Z} \leq \mathbb{Z}$.

4. Cosets

Definition: Let H be a subgroup of the group G. For any arbitrary element $g \in G$, **Left Coset** of H in G containing g, is the set obtained by multiplying every element of H by g on the left:

$$gH = \{g * h \mid h \in H\}$$

the **Right Coset** is the set obtained by multiplying every element of H by g on the right:

$$Hg = \{h * g \mid h \in H\}$$

The element g here is called the **Coset Representative**.

5. Properties of Cosets

Let H be a subgroup, and let a and b be elements of the group G.

- **Property 1 (Absorption): $\mathbf{aH = H \iff a \in H}$**
  Meaning: A coset equals the subgroup itself if and only if its representative is already inside the subgroup.

  Proof. ($\Rightarrow$): Assume $aH = H$. Since the identity $e \in H$, the element $a * e$ must be in aH. Therefore, $a \in aH$. Because $aH = H$, it follows that $a \in H$.
  ($\Leftarrow$): Assume $a \in H$. By closure, for every $h \in H$, $ah \in H$, meaning $aH \subseteq H$. For the converse, let $x \in H$. We can write x as $a(a^{-1}x)$. Since $a \in H$, its inverse $a^{-1} \in H$, and by closure, $a^{-1}x \in H$. Thus, x is the product of a and an element in H, proving $H \subseteq aH$. Therefore, $aH = H$. □

- **Property 2 (The Equality Condition): $\mathbf{aH = bH \iff a^{-1}b \in H}$**
  Meaning: Two elements generate the exact same coset if and only if they differ by an element in H.

  Proof. We can prove this elegantly by reducing it to Property 1. Multiply both sides of the equation $aH = bH$ on the left by a^{-1}:

  $$aH = bH \iff a^{-1}(aH) = a^{-1}(bH) \iff H = (a^{-1}b)H$$

  By Property 1, the coset $(a^{-1}b)H$ is equal to H if and only if the representative $(a^{-1}b)$ is an element of H. Thus, $aH = bH \iff a^{-1}b \in H$. □

- **Property 3 (Strict Partition): Any two cosets of the same subgroup are either perfectly identical or completely disjoint.**

  Proof. Suppose aH and bH are not disjoint. Then, there exists some element x that is in both. That is,

  $$x = ah_1 \text{ and } x = bh_2.$$

  Setting them equal, we get

  $$\begin{aligned} ah_1 &= bh_2 \\ h_1 &= a^{-1}bh_2 \\ h_1h_2^{-1} &= a^{-1}b \end{aligned}$$

  By closure, $h_1h_2^{-1} \in H$. Thus, $a^{-1}b \in H$.

  By Property 2, this implies $aH = bH$.

  Therefore, any two cosets are either perfectly disjoint or exactly identical. □

- **Property 4 (Uniform Size): $|\mathbf{aH}| = |\mathbf{H}|$**
  Meaning: Every coset has the exact same number of elements as the original subgroup.

  Proof. Define a function $f : H \to aH$ by $f(h) = ah$.

  Surjective: Every element in aH is of the form ah, which is mapped to by $f(h)$.

  Injective: If $f(h_1) = f(h_2)$, then $ah_1 = ah_2$. By left cancellation, $h_1 = h_2$.

  The perfect bijection proves $|H| = |aH|$. □

6. Lagrange's Theorem

Lagrange's Theorem: If G is a finite group, and H is a subgroup of G, then $|H|$ divides $|G|$.

Proof. By Property 3, the left cosets of H perfectly partition the group G. This means G is divided into mutually disjoint subsets, leaving no elements behind.

Let $n = |G : H|$ denote the exact number of these distinct cosets (known as the index of H in G). We can write G as the union of these n distinct cosets:

$$G = a_1H \cup ... \cup a_nH$$

Because the cosets perfectly partition G, the order of G is simply the sum of the orders every cosets:

$$|G| = |a_1H| + ... + |a_nH|$$

Then, by Property 4, we know every coset's order is equal to the subgroup itself ($|aH| = |H|$). Therefore, the order of G is just the number of distinct cosets multiplied by the size of the subgroup:

$$|G| = n \times |H|$$

Since n is strictly an integer, this equation demonstrates that $|H|$ perfectly divides $|G|$. □

Notice that we also have the number of distinct left (or right) cosets is just $\frac{|G|}{|H|}$.

7. The Orbit-Stabilizer Theorem

For any given object x:

- **The Orbit (O_x):** The set of all places x can be moved to. ($O_x \subseteq X$)
- **The Stabilizer ($Stab(x)$):** The set of group elements that leave x totally unmoved. ($Stab(x) \leq G$)

Orbit-Stabilizer Theorem: $|G| = |O_x| \times |Stab(x)|$

Proof. By Lagrange's Theorem, $|G| = [G : Stab(x)] \times |Stab(x)|$, where $[G : Stab(x)]$ is the number of distinct cosets of the stabilizer.
We just need to prove that the number of cosets is exactly equal to the size of the orbit $|O_x|$.
Let's map a coset $gStab(x)$ to a destination in the orbit $g \cdot x$.
Is this a 1-to-1 relationship?
$g \cdot x = h \cdot x \iff h^{-1}g \cdot x = x \iff h^{-1}g \in Stab(x) \iff gStab(x) = hStab(x)$.
Yes! Two actions send the object to the exact same destination *if and only if* they belong to the exact same coset of the stabilizer. Thus, the number of distinct destinations ($|O_x|$) equals the number of distinct cosets. Substituting this into Lagrange gives the theorem. □

8. Burnside's Lemma

Burnside's Lemma: The number of distinct orbits $|X/G|$ is equal to the average number of objects fixed by the elements of the group.

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

Proof. We prove this using a classic trick: **Double Counting**. Imagine a list of all pairs (g, x) where action g leaves object x fixed.
Method 1 (Count by actions): Iterate through every group element g, and count how many objects it fixes (the set X^g). Total $= \sum_{g \in G} |X^g|$.
Method 2 (Count by objects): Iterate through every object x, and count how many

actions fix it (the Stabilizer). Total $= \sum_{x \in X} |Stab(x)|$.
Since we counted the same list, we set them equal:

$$\sum_{g \in G} |X^g| = \sum_{x \in X} |Stab(x)|$$

By Orbit-Stabilizer, $|Stab(x)| = \frac{|G|}{|O_x|}$. Substitute this in:

$$\sum_{g \in G} |X^g| = \sum_{x \in X} \frac{|G|}{|O_x|}$$

Factor out $|G|$ and divide:

$$\frac{1}{|G|} \sum_{g \in G} |X^g| = \sum_{x \in X} \frac{1}{|O_x|}$$

Look at the right side. For any single orbit, summing $\frac{1}{|O_x|}$ for every element in that orbit yields exactly 1. Therefore, summing over all objects simply counts the total number of distinct orbits, $|X/G|$. □

9. Example: The 8×8 Chessboard

Suppose we color the 64 squares of a chessboard black and white. How many genuinely unique boards exist if we consider boards identical when rotated *or* flipped (reflected)?

- **The Set (X):** 2^{64} total raw colorings.
- **The Group (G):** The symmetries of a square (the Dihedral group D_4). This includes 4 rotations and 4 reflections. So, $|G| = 8$.

Applying the Lemma (Finding $|X^g|$ for each action): To find the number of boards fixed by an action, we determine how that action groups the 64 squares into independent "orbits" or cycles. If a board is unchanged by an action, all squares within a cycle must be the same color.

1. $0°$ **Rotation (Identity):** Does nothing. All 64 squares are independent. (2^{64} fixed boards)
2. $90°$ **and** $270°$ **Rotations (2 elements):** Squares rotate in cycles of 4. There are $64/4 = 16$ independent cycles. (2^{16} fixed boards each)
3. $180°$ **Rotation:** Squares rotate in cycles of 2 (swapping with their opposites). There are $64/2 = 32$ independent cycles. (2^{32} fixed boards)
4. **Horizontal and Vertical Reflections (2 elements):** Flips the board over the center axes. Because 8 is an even number, no squares lie perfectly on the axis. Every square pairs with one on the opposite side. There are $64/2 = 32$ pairs. (2^{32} fixed boards each)

5. **Diagonal Reflections (2 elements):** Flips the board across the main diagonals. The 8 squares lying exactly on the diagonal remain fixed (8 independent squares). The remaining 56 squares are mirrored in pairs ($56/2 = 28$ pairs). Total independent cycles $= 8 + 28 = 36$. (2^{36} fixed boards each)

The Calculation: Using Burnside's Lemma, we sum these up and divide by $|G|$:

$$\begin{aligned}\text{Unique Boards} &= \frac{1}{8}\left(2^{64} + 2 \cdot 2^{16} + 2^{32} + 2 \cdot 2^{32} + 2 \cdot 2^{36}\right) \\ &= 2,305,843,028,004,192,256 \approx 2.3 \text{ quintillion}\end{aligned}$$